

From Harvest to Food Security: Machine Learning Forecasts for the United States

Cathy Rong^{*1, 2}, Safiya Taher¹, Sabrina Qian¹, Renisa Ghosh¹, and Arthur Jiang¹

¹Old Scona Academic, Edmonton, Alberta, Canada

²United World Colleges Red Cross Nordic, Flekke, Vestland, Norway

January 9, 2026

Abstract

Food insecurity persists even in high-output agricultural systems, underscoring the need for timely national indicators. This study tests whether U.S. agricultural production signals can predict changes in a composite Food Security Index (*FSI*) using annual data (1970-2023) integrating USDA NASS QuickStats measures for three major staple crops: corn, soybeans, and wheat (yield, production, and area harvested) as well as engineering shock and lag features.

The prediction target expresses changes in a *FSI* scaled from $[0, 1]$ (higher = more food secure) constructed from four national indicators, including FAOSTAT supply/production measures and macro-climate variables: Food production index, Dietary energy supply, GDP per capita, and Temperature change on land. We first used regression models (Random Forest, RidgeCV, and Gradient Boosting) trained on 1970-2012 and tested on 2013-2023 data to predict annual changes in food security, ($\Delta FSI(t)$), evaluating accuracy with RMSE/MAE/ R^2 against a persistence baseline $FSI(t - 1)$, and then converted (ΔFSI) into a binary direction label (positive (> 0), negative (≤ 0)) to assess how well the same models predict year-to-year improvement vs decline using balanced accuracy.

Overall, results showed that the national crop statistics contain measurable predictive indicators for annual food security dynamics and motive extensions.

Keywords:

Agriculture, Food Security Index, Machine Learning

1 Introduction

Food insecurity persists even in highly productive agricultural systems because national food security depends on more than total crop output [1]. It also depends on whether people can reliably access safe, sufficient, nutritious food and whether the system remains resilient to shocks [2]. With climate variability [3], market disruptions [4], and economic pressures increasingly interacting, there is growing demand for timely, interpretable national indicators that can be tracked consistently over long periods to support decision making [5].

Agricultural production is among the most consistently measured national variables. In the United States, corn, soybeans, and wheat are especially well documented because they dominate staple crop production and shape downstream food and feed markets [6, 7]. Annual changes in yields, acreage, and output can reflect both biophysical constraints and land-use responses. However, production alone does not map cleanly onto national food security outcomes: access and affordability [8], supply chains [9, 10], and broader macro conditions introduce nonlinearities and time-lagged effects.

Researchers have developed multiple food-security indicators to reflect food security’s multifaceted nature. The Global Food Security Index (GFSI) [11, 12], for example, assesses 113 countries using measures of affordability, availability, quality and safety, and sustainability/adaptation, offering a defensible, simple, and transparent basis for agricultural policymaking [13, 14].

In this study, a composite Food Security Index (*FSI*) is developed for US agricultural system using four Food and Agriculture Organization of the UN’s corporate statistical database (FAO-

^{*}Corresponding author: cathyrong11@gmail.com

Word Count (excluding figures, tables, references, and title page): 4079

STAT) indicators [15]: the food production index, dietary energy supply, GDP per capita, and temperature change on land. This macro summary score tracks year-to-year national food security with long-run consistent data and serves as a proxy for overall conditions (not household hunger). The production index indicates domestic supply availability [16], dietary energy supply reflects dietary adequacy (quality) [17], GDP per capita captures economic access for affordability [18], and temperature change on land represents sustainability in a period [19].

2 Materials & Methods

2.1 Data and Study Design

We assessed whether U.S. staple-crop production predicts changes in a composite Food Security Index (*FSI*) using an annual time series from 1970-2023 [20, 21]. Crop predictors were drawn from United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) QuickStats for corn, soybeans, and wheat, including yield, total production, and harvested area for each crop (N=1,944) [22]. The response variable was defined as a change in *FSI* scaled to [0, 1], with higher values indicating better food security.

In addition to crop exclusive statistics, we included exogenous drivers: a food Consumer Price Index (CPI) and its annual change, World Bank “Pink Sheet” indices (Energy, Agriculture, Food, Grains, Fertilizers, Metals & Minerals) and selected commodity prices (crude oil, maize, soybeans, and wheat—Hard Red Winter and Soft Red Winter) [15]. All datasets were restricted to the U.S., aggregated to annual values, and merged by year into a master table; non-numeric entries were treated as missing and dropped only when required for model fitting to preserve predictor-target alignment. The resulting dataset was then used for feature engineering (lags and year-over-year “shock” variables) and model evaluation using time-ordered train/test splits.

Figure 1 summarizes the end-to-end flow (data integration through validation). To reduce bias and prevent leakage, we excluded state-level QuickStats features, ensuring model development and evaluation reflected the national U.S. context.

2.2 Data Preparation

Crop predictors were retrieved via the USDA NASS QuickStats API [22] by querying U.S.-level annual records for corn, soybeans, and

wheat and extracting yield, production, and harvested area. API calls were executed in Python using the `requests` library, and each metric was saved as a structured CSV. Exogenous series were obtained from the U.S. Bureau of Labor Statistics (food CPI) [23], the World Bank Commodity Price Data (“Pink Sheet”) for indices and commodity prices [24, 25], and World Bank population totals.

All preprocessing was performed in Python [26] using `pandas` [27] and `NumPy` [28] for parsing, numeric conversion, reshaping, merging, and feature engineering (lag and shock variables). Non-numeric entries were treated as missing and removed only when required for model fitting; `pathlib` supported file handling [29], and `scikit-learn` utilities supported model-ready preprocessing [30]. Cleaned series were aggregated to one row per year and merged with the constructed *FSI* and exogenous predictors to form the final 1970-2023 modeling dataset. Continuous features were standardized using `scikit-learn`’s `StandardScaler` to improve comparability across units and optimize model performance.

2.3 Targets and Feature Engineering

We construct an annual composite Food Security Index, $FSI(t) \in [0, 1]$, where higher values indicate better food security. The index is based on four national indicators:

1. $F(t)$: Food production index
2. $K(t)$: Dietary energy supply (kcal per capita per day)
3. $G(t)$: GDP per capita (USD)
4. $P(t)$: Temperature change on land ($^{\circ}\text{C}$)

To make components comparable, each series is min-max normalized to [0, 1] using the 1970-2012 training period:

$$X_{\text{norm}}(t) = \frac{X(t) - \min(X)}{\max(X) - \min(X)}$$

for

$$X \in \{F, K, G, P\}$$

Each component lies in a range of [0,1] after this.

Secondly, we combined all four components into one index by geometrical mean. This is because we try to avoid one pillar masking another and penalize low scores.

$$FSI(t) =$$

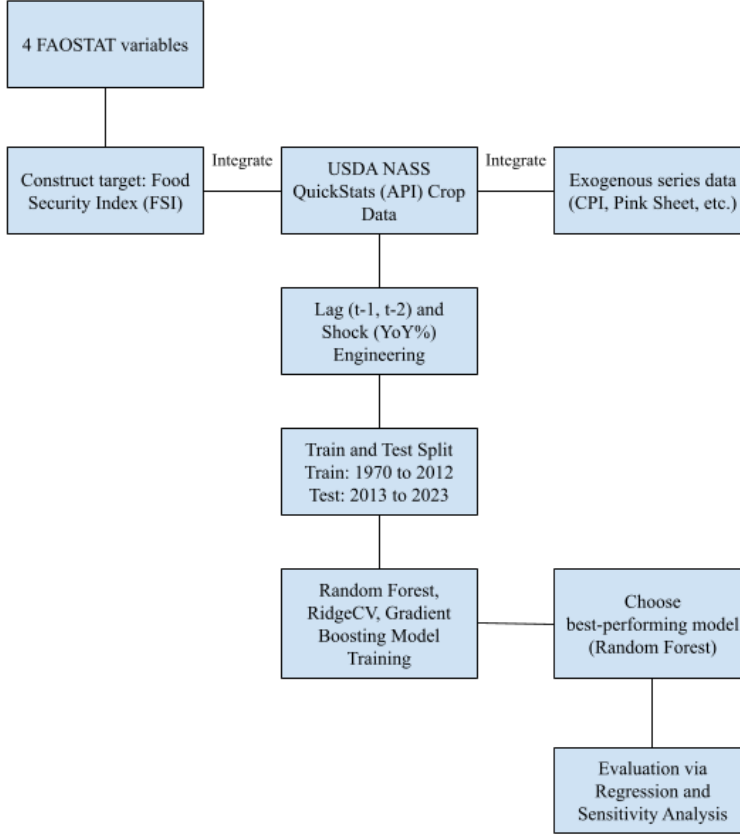


Figure 1: Study Flow Chart

$$\sqrt[4]{(K_{\text{norm}}(t) * P_{\text{norm}}(t) * G_{\text{norm}}(t) * F_{\text{norm}}(t))}$$

This, in turn, makes $FSI(t)$ a value within $[0,1]$, where a higher FSI value corresponds to stronger food security conditions, by our composite definition. At the same time, a drop in FSI would signify a deterioration driven by one or more components declining in relevance to their historical range.

Although FSI is interpretable at annual resolution, its strong persistence over time reflects forecasting year-to-year rather than its absolute levels. Therefore, we derived an indicator for FSI changes to classify improvements (positive) versus deteriorations (negative).

$$\Delta FSI(t) = FSI(t) - FSI(t-1)\%$$

Once $\Delta FSI(t)$ is negative, indicating a decline in food security, policy intervention may be needed to restore a stable or improving trajectory.

For prediction, we use crop indicators (yield, production, harvested area for corn, soybeans, and wheat) and exogenous macro variables. To capture disruptions and delayed effects, we engineer (i) yearly shock features (year-over-year changes; percent changes for strictly positive

variables) and (ii) 1-2 year lags of both levels and shocks, allowing the model to learn immediate versus lagged associations with ΔFSI .

2.4 Model Selection

To assess whether the predictors contain usable signal for national food security, we evaluated multiple supervised regression models to predict ΔFSI . Models were trained using a time-ordered split (1970-2012) and tested on 2013-2023, and performance was benchmarked against a persistence baseline that forecasts the next year's value.

We compared a regularized linear Ridge regression and a non-linear Random Forest ensemble. Ridge provides a transparent benchmark that mitigates multicollinearity via regularization, whereas Random Forest captures nonlinearities and interactions without imposing a parametric functional form. Across change-based (ΔFSI) formulations, we selected the best approach based on predictive accuracy on the held-out test period.

2.5 Performance Metrics

Predictive accuracy of level formulation ΔFSI along time was measured using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 values, reported in relevance to the persistence baseline. For the change-based formulation, we assessed directional performance additionally by classifying whether ΔFSI was positive and then computing a confusion matrix and balanced accuracy along with it. Receiver Operating Characteristic Area Under the Curve (ROC AUC) and average precision were also calculated using the changed, predicted indicator.

3 Results

Using national staple-crop statistics and exogenous economic indicators, we first evaluated three different model configurations, Random Foresting, RidgeCV, and GradientBoosting, to predict ΔFSI with regression. Across all model configurations FSI showed strong yearly persistence, meaning a "no-change" forecast, anchored on the persistence baseline (prior year level), already achieved strong predictive performance.

Our second step in our project was to predict the direction of change for our ΔFSI . We first converted ΔFSI into a binary direction label ($\Delta FSI > 0$ vs $\Delta FSI \leq 0$) to demonstrate cleared separation in performance with our same model set.

Figure 2 visualizes regression results from our Random Forest model. We translated out ΔFSI back into levels by reconstructing predicted food security as: $FSI_{pred}(t) = FSI(t-1) + \Delta FSI_{pred}(t)$.

Over the held-out test window, this plot contrasts reconstructed predictions with observed FSI. The regression model captures significant year-to-year signal from the predictor set, as evidenced by the reconstructed series' broad trajectory of the observed index in later years. However, the biggest deviations happen during sharper transition periods, when the observed series changes quickly in comparison to its typical persistence. Predicted values in these pivotal years differ significantly from observed levels, indicating that crop indicators and engineered macroclimate features alone may not be sufficient to explain sudden regime shifts. The predicted series continues to be below the observed series for several years after this transition period, indicating a propensity to underestimate food security during long-term recovery or stabilization. Overall, Figure 2 indicates that production-informed predictors con-

tain measurable temporal signal, but that reconstruction accuracy is constrained primarily by turning points rather than by stable years.

Figure 3 shows residuals computed as Actual - Predicted, where positive residuals indicate underprediction (Observed $FSI > \text{actual}$) and negative residuals indicate overprediction. This is defined by the equation: $r(t) = FSI_{actual}(t) - FSI_{pred}(t)$. The residuals show that the model's biggest errors occur in a small subset of years rather than being evenly distributed, ranging from roughly -0.50 at the largest overprediction (early in the test window) to roughly $+0.66$ at the largest underprediction (mid-window). The model exhibits a significant overprediction (≈ -0.50) after a small positive residual in the first year ($\approx +0.05$), which is followed by a series of large positive residuals during a transition period ($\approx +0.32, +0.66$, and $+0.49$ across consecutive years). This pattern, which alternates between overshooting and persistent undershooting as the observed series changes regime, indicates that the model is unable to track a sudden change in the underlying trajectory. In later years, residuals are consistently small and positive ($\approx 0.10 - 0.25$), indicating that errors concentrate around turning points and the model tends to underpredict thereafter rather than fully catching up.

The importance of features for predicting FSI was evaluated based on the model coefficients. Features with larger absolute coefficient values were deemed more significant, indicating a stronger influence on the target outcome.

Figure 4 shows the top 15 feature importances from predicting ΔFSI . Two climate-related predictors, temp_change_c (≈ 0.245) and t_norm (≈ 0.23), have a high concentration of importance. Together, they account for approximately 0.47 to 0.48 of the total model importance, or nearly half of the explained split gain in the forest. The next most significant predictors, MAIZE_lag2 and soybeans_area_harvested_yoy (each ≈ 0.05), form a second tier at significantly smaller magnitudes. Macro/commodity and crop-shock terms like CRUDE_PETRO_yoy_pct (≈ 0.03) and corn_yield_derived_yoy ($\approx 0.02-0.025$) are included in a third tier. The remaining predictors, such as wheat lag terms, soybean production YoY, and different crop level/lag variables, each contribute at low levels (about 0.01-0.02), creating a long tail of smaller effects.

Table 1 shows how sensitive the Random Forest ΔFSI model is to the way we encoded short-run disruptions and delayed effects in our predictors. We varied two feature types: our shock features (year by year percentage changes, "_-

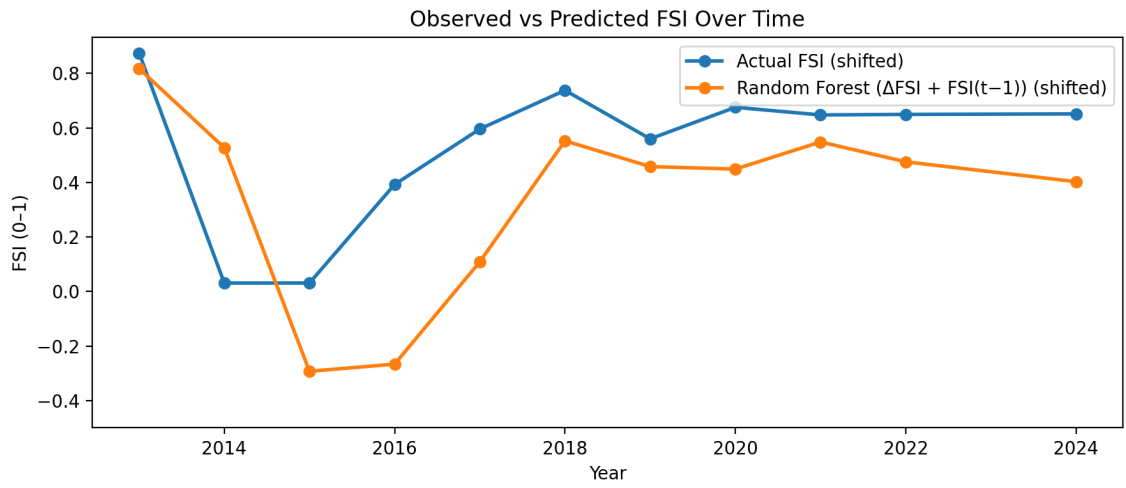


Figure 2: Observed vs Predicted FSI Over Time: Random Forest vs Persistence (2013-2023)

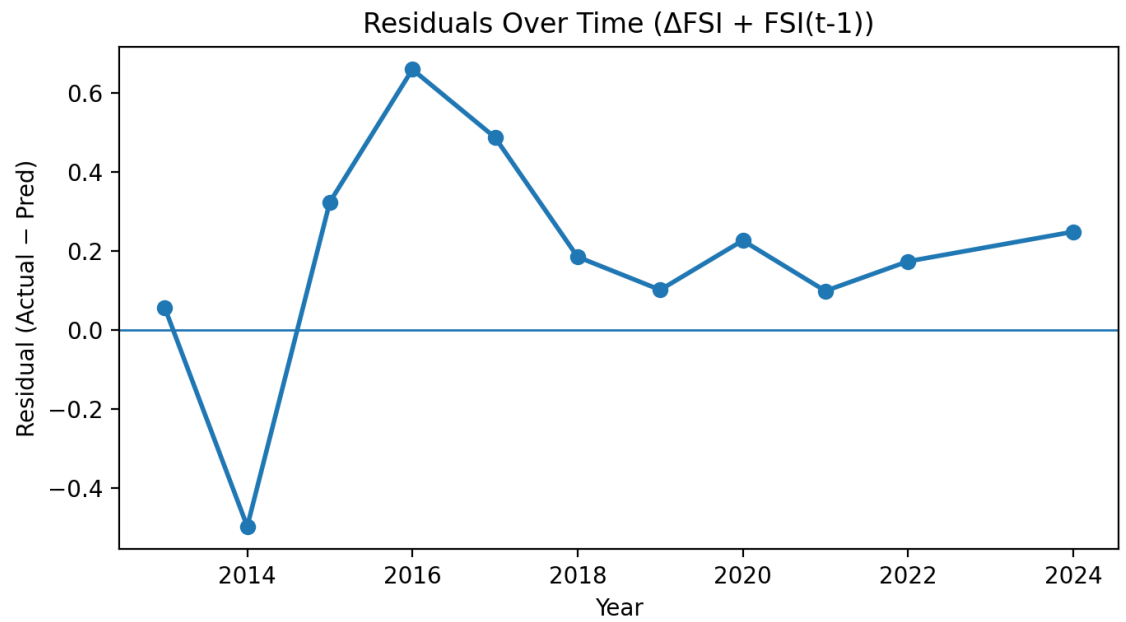


Figure 3: Residuals Over Time (Observed - Predicted FSI), 2013-2023

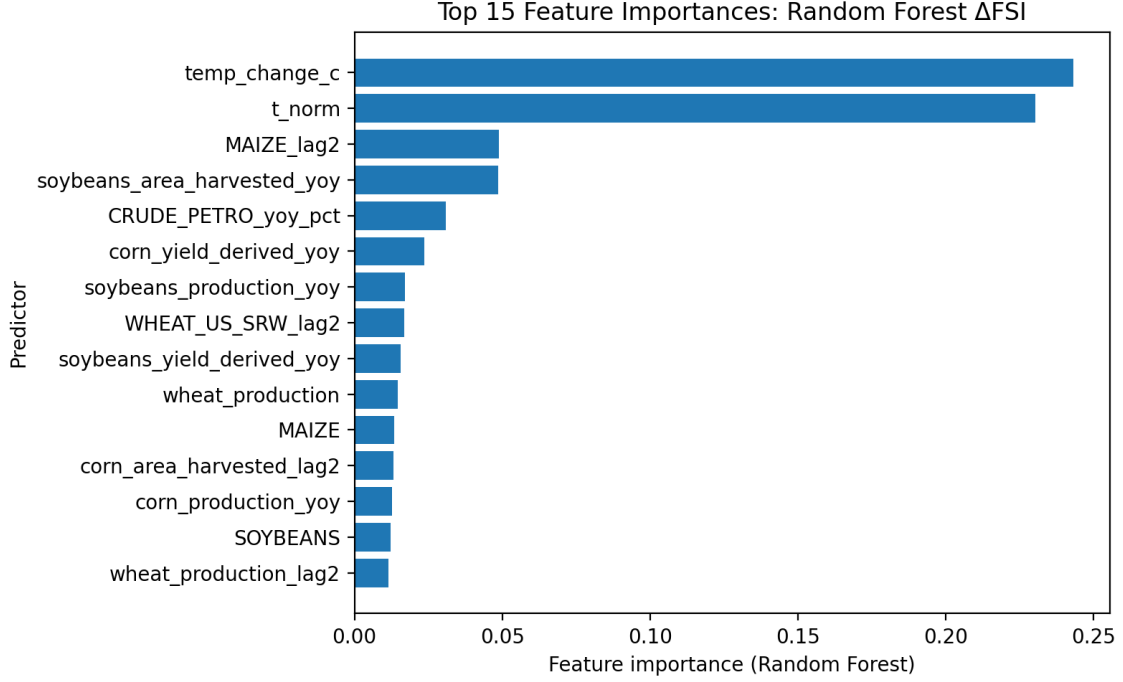


Figure 4: Top 15 Feature Importances: Random Forest ΔFSI

yoy_pct”) and our lag features (prior year values, “_lag1” or “_lag2”). This setup helps isolate whether the model benefits more from immediate year by year changes, historical memory, or both.

The persistence baseline $FSI(t) = FSI(t-1)$ is already strong (RMSE = 0.0669, MAE = 0.0531, $R^2 = 0.6727$), highlighting substantial year-to-year stability in the index. Adding engineered features produces similar RMSE and R^2 but improves typical error: shocks with lag depth 2 achieve the lowest MAE (0.0476–0.0487) with RMSE remaining near 0.0671–0.0674 and R^2 near 0.667–0.670, while the lag1-only variant is slightly weaker (RMSE = 0.0682, MAE = 0.0495, $R^2 = 0.6586$). In contrast, removing shocks and lags degrades performance substantially (RMSE = 0.0755–0.0775, MAE = 0.0567–0.0680, $R^2 = 0.5458$ –0.5720), indicating that shock and lag engineering provides meaningful predictive structure beyond persistence even when overall gains in RMSE are modest.

Figure 5 is an evaluation on whether the model is able to predict the direction of food security change on an annual scale by framing the outcome as a binary label. Our definitions included: Positive $\Delta FSI > 0$ and Negative $\Delta FSI \leq 0$. This figure includes two visualizations. The ROC curve (5a) shows a moderate level of discriminative ability, with an AUC of 0.71. This means that the model shows significant overlap between classes while ranking improvement years above non-improvement

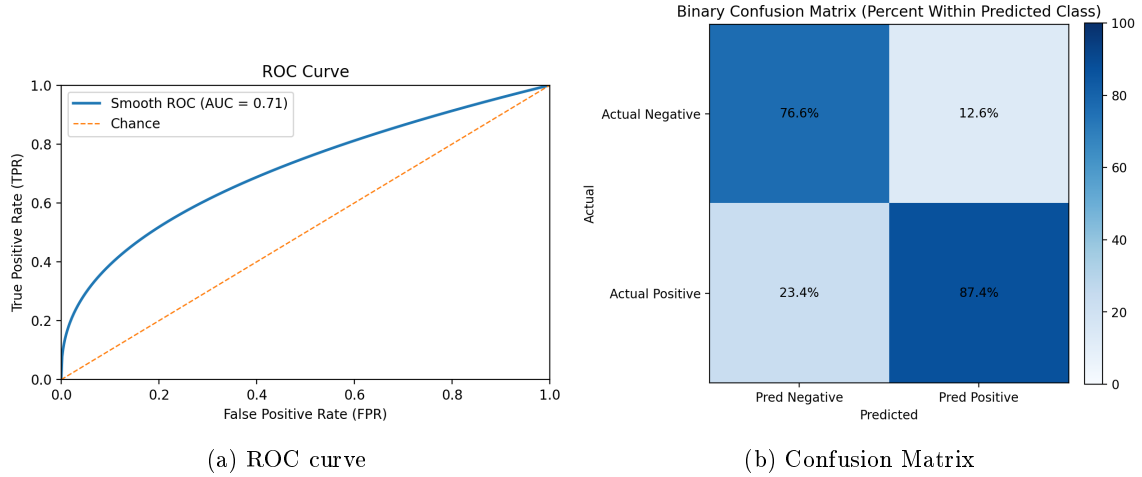
years significantly better than chance. Our second visualization (5b) is the confusion matrix which is shown as percentages with the predicted classes (normalized by column), so each column is a reflection of how accurate the model’s predicted label is. When the model predicts Positive, 87.4% of those predictions correspond to true improvement years (true positives), while 12.6% correspond to false alarms (false positives). In contrast, when the model predicts Negative, 76.6% of those predictions are correct non-improvement years (true negatives), but 23.4% are missed improvement years (false negatives).

Table 2 is a comparison of directional performance across a full set of models by evaluating whether each approach correctly predicts the sign of annual food security change, where Positive is $\Delta FSI > 0$ and Negative is ≤ 0 . While the earlier results focus exclusively on the Random Forest ΔFSI model for detailed diagnostics, Table 2 is a showcase of the other models we tried to test with by contextualizing how alternative methods and feature sets perform on the same 2013 to 2023 test window.

The best-performing model is the Random Forest using the full feature set (crops+macro+shocks+lags), achieving balanced accuracy = 0.8391 with high sensitivity (0.8949) and strong specificity (0.7831), indicating reliable identification of both improvement and non-improvement years. Removing crop information reduces discrimination: the Ran-

Table 1: Shock-Lag Sensitivity Analysis for the Random Forest ΔFSI Model (Test: 2013-2023)

Configuration	Shocks	Lag Depth	Lagged Shocks	RMSE	MAE	R ²	Balanced Accuracy
Persistence baseline (FSI(t)=FSI(t-1))	—	—	—	0.0699	0.0554	0.6443	
Shocks + lag1+lag2 (no lagged shocks)	On	2	Off	0.0701	0.0509	0.6419	0.7143
Shocks + lag1+lag2	On	2	On	0.0704	0.0497	0.6393	0.7143
Shocks + lag1	On	1	On	0.0712	0.0517	0.6308	0.6429
No shocks + no lags	Off	0	Off	0.0788	0.0646	0.5478	0.5
Shocks only	On	0	Off	0.0791	0.0592	0.5438	0.6429
Lag1 only	Off	1	Off	0.0809	0.071	0.5227	0.5


Figure 5: ROC Curve and Confusion Matrix for Predicting $\Delta FSI > 0$

dom Forest with macro+shocks+lags drops to balanced accuracy = 0.7644 (sensitivity = 0.7458, specificity = 0.7831), suggesting that crop indicators provide additional signal beyond macro-climate shocks alone. Gradient Boosting exhibits an imbalanced error profile (balanced accuracy = 0.7085) with high sensitivity (0.8949) but lower specificity (0.5221), while RidgeCV further amplifies this trade-off by predicting positives aggressively (sensitivity = 1.0000) at the cost of very low specificity (0.2610) and lower balanced accuracy (0.6526). The persistence baseline performs near chance (balanced accuracy = 0.5221), underscoring that directional improvements are not explained by persistence alone and are captured most effectively by the full-feature Random Forest.

4 Discussion

Overall, our results show that national crop yield, production, and harvest-area statistics

can help track U.S. food security dynamics over time. The Random Forest ΔFSI framework provided the most reliable performance among the tested approaches, demonstrating that supervised models can learn relationships between staple-crop indicators and a composite food security index using annual U.S.-only data (1970–2023), particularly when the target is modeled as year-to-year change (ΔFSI) rather than absolute levels. In the 2013–2023 holdout period, the reconstructed predictions generally followed the observed trajectory and performed competitively against a strong persistence benchmark, supporting the notion that national crop conditions correspond to broader food-system outcomes and can contribute to monitoring and forecasting.

A key takeaway is that shock and lag feature engineering mattered nearly as much as model choice. The shock-lag sensitivity results show that removing both YoY shock terms and lag structure worsened performance, whereas in-

Table 2: Directional Model Comparison for ΔFSI :
Balanced Accuracy, Sensitivity, and Specificity (Test: 2013-2023)

Setup	Model	Balanced Accuracy	Sensitivity	Specificity
ΔFSI : crops+macro+shocks+lags	RandomForest	0.8036	0.8571	0.75
ΔFSI : macro+shocks+lags	RandomForest	0.7321	0.7143	0.75
ΔFSI : crops+shocks+lags	GradientBoosting	0.6786	0.8571	0.50
ΔFSI : crops+shocks+lags	RidgeCV	0.625	1	0.25
Baseline	Persistence ($\Delta = 0$)	0.5	0	1

cluding both consistently improved results. This is plausible in food systems where production outcomes, input costs, and price conditions influence availability and stability with delays rather than instantaneously. Consistent with this mechanism, lagged crop quantities (e.g., $t-1$, $t-2$) and year-over-year shock terms (YoY %) appear among the strongest predictors in Figure 4, suggesting that the model benefits from both the “memory” of prior conditions and the detection of sudden disruptions.

At the same time, our results clarify what production-centered prediction can and cannot do. The residual analysis shows that the largest errors occur around sharper transitions, indicating that turning points are harder to reconstruct from crop and engineered macro-climate features. Directional evaluation of up/down changes in ΔFSI shows moderate discrimination ($AUC \approx 0.71$), which is consistent with food security not being determined by production alone; affordability, distribution, macroeconomic shocks, and policy/trade dynamics can shift outcomes even when harvest data are strong. As a result, production and commodity indicators are most useful for identifying medium-term trends and gradual movement, and less trustworthy when access or policy forces dominate the signal.

Several factors also limit directional performance. First, the test window spans only 11 years, so a few borderline years can noticeably shift the ROC curve. Second, the positive/negative boundary is susceptible to small noise in either the predictors or the constructed FSI , because many national year-to-year changes are small and near $\Delta FSI = 0$. Third, even when agricultural production contains signal, the sign of year-to-year change depends on drivers only partially represented in our features (e.g., policy shifts, trade dynamics, household prices, and access constraints). For these reasons, the ROC/AUC results should be interpreted as a supporting diagnostic indicating some ability to detect improvement years rather

than a definitive early-warning classifier.

A further limitation concerns the binary directional definition itself (Positive if $\Delta FSI > 0$, Negative if $\Delta FSI \leq 0$). The sign alone can overstate practical significance: for example, if FSI rises from 0.60 to 1.00 and then decreases slightly to 0.99, the final year is labeled “Negative” despite remaining substantially more food secure than the initial year. Future work could introduce a magnitude-based warning rule (e.g., only flagging declines when the relative drop exceeds a threshold such as $\% \Delta FSI \leq -10\%$) or a neutral band around zero to separate trivial fluctuations from policy-relevant deterioration.

Additional limitations should be considered. Localized shocks that do not register in aggregates may be missed by the national annual design, which smooths within-country variation. Although FSI provides a consistent 0–1 target, any index depends on design choices that can affect year-to-year dynamics. Despite including several macro and commodity-price series, the feature set omits important determinants such as household income and poverty, policy and assistance changes, trade volumes, inventories/stock-to-use conditions, and distribution constraints.

Otherwise, our findings are directly relevant to policy. First, they support using routinely available crop statistics as part of an early-warning style monitoring toolkit, particularly for detecting deviations from recent norms when using YoY shocks and short lags. This can help agencies prioritize attention, scenario planning, and preparedness. Second, the limitations are equally informative: moderate ROC/AUC performance and outlier years reinforce that supply-side strength alone does not guarantee improved food security, implying that yield-boosting or acreage-expansion policies should be paired with measures targeting affordability and access (e.g., price stabilization, safety-net strengthening, and distribution reliability).

In conclusion, crop indicators alone cannot fully explain food security, but they are suffi-

ciently informative to support national tracking and monitoring, especially when modeled with shocks and lags. The clearest path to more policy-relevant forecasting is expanding predictors toward access, trade, and policy drivers, and treating year-to-year changes as more predictable when models can learn from both short-run disruptions and delayed effects.

5 Conclusion

Even in a high-producing agricultural system, such as the United States, national food security can also shift, which makes timely monitoring and early warnings especially important. In conclusion, crop indicators by themselves cannot fully explain food security, but they are sufficiently informative to support tracking and monitoring at the national level, particularly when modelled with shocks and lags. The most compelling evidence from this work is that expanding the predictor set toward access, trade, and policy drivers is the most promising route to policy-relevant forecasting, and that year to year changes in food security are more predictable when models can learn from both short-run disruptions and delayed effects. Future work should focus on further increasing our predictor set to include more variables such as access, trade, and policy drivers and move toward uncertainty-aware forecasts. That allows for decision-makers to more effectively allocate resources and respond earlier to conditions that threaten national food security. This approach has the potential to respond to noticed issues faster and more effectively identifying food insecurity risk.

Acknowledgements

The team would like to thank the support from Dr. Jianshan Chen in University of Alberta.

References

- [1] Michael A. Long, Lara Gonçalves, Paul B. Stretesky, and Margaret Anne Defeyter. Food insecurity in advanced capitalist nations: A review. *Sustainability*, 12(9), 2020.
- [2] Aikaterini Kavallari, Thomas Fellmann, and S. Hubertus Gay. Shocks in economic growth = shocking effects for food security? *Food Security*, 6(4):567–583, August 2014.
- [3] Robert H. Beach, Allison M. Thomson, and Bruce A. McCarl. Climate change impacts on us agriculture. *IATRC Public Trade Policy Research and Analysis Symposium*, 2010. Available from: <https://ageconsearch.umn.edu/record/91393>.
- [4] Regis Barnichon, Christian Matthes, and Alexander Ziegenbein. Are the effects of financial market disruptions big or small? *The Review of Economics and Statistics*, 104(3):557–570, 05 2022.
- [5] J. L. Hatfield, K. J. Boote, B. A. Kimball, L. H. Ziska, R. C. Izaurralde, D. Ort, A. M. Thomson, and D. Wolfe. Climate impacts on agriculture: Implications for crop production. *Agronomy Journal*, 103(2):351–370, 2011.
- [6] Jasmien De Winne and Gert Peersman. Macroeconomic effects of disruptions in global food commodity markets: Evidence for the united states. *Brookings Papers on Economic Activity*, 2016(2):183–286, 2016. Fall 2016.
- [7] Maayan Kreitzman, Eric Toensmeier, Kai M. A. Chan, Sean Smukler, and Navin Ramankutty. Perennial staple crops: Yields, distribution, and nutrition in the global food system. *Frontiers in Sustainable Food Systems*, Volume 4 - 2020, 2020.
- [8] N. M. A. Manap and N. W. Ismail. Food security and economic growth. *International Journal of Modern Trends in Social Sciences*, 2(8):108–118, 2019.
- [9] Noé J. Nava, William Ridley, and Sandy Dall’erba. A model of the u.s. food system: What are the determinants of the state vulnerabilities to production shocks and supply chain disruptions? *Agricultural Economics*, 54(1):95–109, 2023.
- [10] B. Gail Smith. Developing sustainable food supply chains. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 363(1492):849–861, 08 2007.
- [11] Economist Impact Expert panel. Global food security index 2022: Exploring challenges and developing solutions for food security across 113 countries, 2022. Available from: <https://impact.economist.com/sustainability/project/food-security-index/>.
- [12] Akinniyi Akinwande. Nigeria agri-econ indicators (1950-2023), 2024. Available from: <https://www.kaggle.com/datasets/akinnyiakwande/nigeria-agri-econ-indicators-1950-2023>.

- [13] Farhana Bibi and Azizur Rahman. An overview of climate change impacts on agriculture and their mitigation strategies. *Agriculture*, 13(8), 2023.
- [14] Rosana Salles-Costa, Aline Alves Ferreira, Ruben Araujo de Mattos, Michael E. Reichenheim, Rafael Pérez-Escamilla, Juliana de Bem-Lignani, and Ana Maria Segall-Corrêa. National trends and disparities in severe food insecurity in brazil between 2004 and 2018. *Current Developments in Nutrition*, 6(4), Apr 2022.
- [15] FAOSTAT. Country indicators, 2025. Available from: <https://www.fao.org/faostat/en/#country>.
- [16] M Nubé. Confronting dietary energy supply with anthropometry in the assessment of undernutrition prevalence at the level of countries. *World Development*, 29(7):1275–1289, 2001.
- [17] Maeve Henchion, Maria Hayes, Anne Maria Mullen, Mark Fenelon, and Brijesh Tiwari. Future protein supply and demand: Strategies and factors influencing a sustainable equilibrium. *Foods*, 6(7), 2017.
- [18] Peter Warr. Food insecurity and its determinants. *Australian Journal of Agricultural and Resource Economics*, 58(4):519–537, 2014.
- [19] İbrahim Bozkurt and MUHAMMED VEYSEL Kaya. Agricultural production index: International comparison. *Agricultural Economics (Czech Republic)*, pages 236–245, 2021.
- [20] Justin Oh. Total emissions per country (2000-2020), 2023. Available from: <https://www.kaggle.com/dsv/5097715>.
- [21] Hannah Ritchie, Pablo Rosado, and Max Roser. Agricultural production. *Our World in Data*, 2023. Available from: <https://ourworldindata.org/agricultural-production>.
- [22] U.S. Department of Agriculture, National Agricultural Statistics Service. Quick Stats Agricultural Database, 2025. Accessed via the Quick Stats API: <https://quickstats.nass.usda.gov/api>.
- [23] Varshita Nalluri. Crop price prediction dataset, 2024. Available from: <https://www.kaggle.com/datasets/varshitanalluri/crop-price-prediction-dataset>.
- [24] World Bank Group and partners. World development indicators - databank, 2015. Available from: <https://databank.worldbank.org/id/db4738db>.
- [25] World Bank Group and partners. World bank solutions to food insecurity - food security update, 2025. Available from: <https://www.worldbank.org/en/topic/agriculture/brief/food-security-update>.
- [26] Python Software Foundation. *Python Language Reference, version 3.11*, Oct 2024. <https://docs.python.org/3/reference/index.html>.
- [27] The pandas development team. pandas-dev/pandas: Pandas, February 2020.
- [28] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, and Ralf Gommers et. al. Array programming with NumPy. *Nature*, 585(7825):357–362, September 2020.
- [29] Python Software Foundation. The Python™ Programming Language. Documentation 3.11, 2026. Specifically for the pathlib module.
- [30] F. Pedregosa, Gaël Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in python. *Journal of machine learning research*, 12(Oct):2825–2830, 2011.